PRACTICE QUESTIONS

DISCRETE MATHEMATICS AND GRAPH THEORY – BE04000261

MODULE – 2

SET THEORY

Inclusion - Exclusion Principle

1 In a class of 35 students, 24 like to play cricket and 16 like to play football. Also, each student likes to play at least one of the two games. How many students like to play both cricket and football?

2 There are 200 individuals with a skin disorder, 120 had been exposed to the chemical C_1, 50 to chemical C_2 and 30 to both the chemicals C_1 and C_2. Find the number of individuals exposed to (i) chemical C_1 but not chemical C_2 (ii) chemical C_2 but not chemical C_1 (iii) chemical C_1 or chemical C_2.

3 How many integers between 1 and 1000 are **not** divisible by 2, 3, or 5?

4 In a survey of 60 people, it was found that 25 people read newspaper H, 26 read newspaper T, 26 read newspaper I, 9 read both H and I, 11 read both H and T, 8 read both T and I, 3 read all three newspapers. Find: (i) the number of people who read at least one of the newspapers. (ii) the number of people who read exactly one newspaper.

5 A college awarded 38 medals in football, 15 in basketball and 20 in cricket. If these medals went to a total of 58 men and only three men got medals in all the three sports, how many received medals in exactly two of the three sports?

COUNTING AND COMBINATORICS

Combinations without repetition (with restrictions)

1 A group of 6 persons is to be formed from 7 men and 5 women. In how many ways can this be done when (i) at least 2 women are included (ii) at most 3 women are included?

2 A group consists of 7 boys and 5 girls. Find the number n of ways to choose: (i) one group leader (ii) two representatives, one boy and one girl (iii) one president and one secretary.

3 A cricket squad has 15 players, including 3 wicket keepers and 5 bowlers. In how many ways can a team of 11 players be selected if exactly 1 wicket keeper and at least 3 bowlers must be selected?

4 Suppose repetitions are not allowed, then
(i) how many four-digit numbers can be formed from six digits 1, 2, 3, 5, 7 and 8?
(ii) how many of such numbers less than 4000?
(iii) how many in (i) are even?
(iv) how many in (i) are divisible by 10?

5 A project team will be formed from 8 engineers and 5 designers. How many ways can a team of 5 people be selected if (i) at most 2 designers (ii) at least 3 engineers are included?

MODULE – 3

FUNCTIONS

Composition of two or three functions

1 If $f(x) = 2x$, $g(x) = x^2$ and $h(x) = x + 1$ then find (i) $f \circ (g \circ h)$ (ii) $(f \circ g) \circ h$.

2 If $f(x) = \sqrt{9 - x^2}$ and $g(x) = \sqrt{9 - x}$ then find $f \circ g(x)$ and $g \circ f(x)$.

3 If $f(x) = \frac{x+1}{x-1}$ and $g(x) = \frac{x-1}{x+1}$ then find $g \circ f(x)$ and $f \circ g(x)$.

4 Let $f(x) = x^2 + 3$, $g(x) = \sin(x)$ and $h(x) = 2x - 1$. Then, find (i) $f \circ (g \circ h)$ (ii) $(f \circ g) \circ h$.

5 Define bijective function. Show that the functions $f(x) = 2x + 5$ and $g(x) = \frac{x-5}{2}$ for $x \in \mathbb{R}$, are inverses of each other.

MODULE – 4

RELATIONS

Equivalence relation

1 Define equivalence relation. Let $A = \{1, 2, 3, 4\}$ and $R = \{(1,1), (1,4), (4,1), (4,4), (2,2), (2,3), (3,2), (3,3)\}$ be the relation on A. Determine whether the relation is an equivalence.

2 Let $X = \{1, 2, 3, 4, 5, 6, 7\}$ and $R = \{(x, y) | x - y \text{ is divisible by } 2\}$. Determine whether the relation is an equivalence.

3 Let $\mathbb{Z}$ be the set of integers and aRb if and only if $a - b$ is divisible by 5 where $a, b \in \mathbb{Z}$. Determine whether the relation is an equivalence.

4 Given that $(a, b) \in R$ if and only if $a = b$ where $a \in A$, $b \in A$ and $A = \{2, 4, 5, 7\}$. Determine whether the relation is an equivalence.

5 If R is the relation defined on $\mathbb{Q}$ (the set of rational numbers) as $R = \left\{\left(\frac{a}{b}, \frac{c}{d}\right) \in \mathbb{Q} \times \mathbb{Q} : ad = bc\right\}$ then show that R is an equivalence relation.

Warshall's Algorithm to find transitive closure of a relation

1 Let $R = \{(1,2), (2,3), (3,3)\}$ be the relation on $A = \{1,2,3\}$. Find the transitive closure of R using Warshall's algorithm.

2 Let $R = \{(1,2), (2,3), (1,3)\}$ be the relation on $A = \{1,2,3\}$. Find the transitive closure of R using Warshall's algorithm.

3 Use Warshall's Algorithm to find the transitive closure of the relation $R = \{(1,4) , (2,1) , (2,3) , (3,1) , (3,4), (4,3)\}$ on $\{1, 2, 3, 4\}$.

4 Let $A = \{1, 2, 3, 4\}$. For the relation R whose matrix is given, find the matrix of the transitive closure by using Warshall's algorithm.

$$M_R = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

5 Find the transitive closure by Warshall's algorithm if $A = \{1, 2, 3, 4, 5\}$ and $R = \{(1,4), (2,1), (2,5), (2,4), (4,3), (5,3), (3,2)\}$.

PARTIAL ORDERING

Hasse Diagram

1 Draw the Hasse diagram for the partial ordering $\{(A, B) \mid A \subseteq B\}$ on the power set $P(S)$ where $S = \{a, b, c\}$.

2 Let A be the finite set and $\rho(A)$ be the power set of A. Let $\subseteq$ be the inclusion relation on the elements of $\rho(A)$. Draw the Hasse diagram of $\langle \rho(A), \subseteq \rangle$ for $A = \{1, 2, 3\}$.

3 Consider a poset $(\{3, 5, 9, 15, 24, 45\}, \mid)$ where $\mid$ denotes "divides". Draw its Hasse diagram.

4 Let $A = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 12, 18, 24\}$ be ordered by the relation x divides y. Draw the Hasse diagram.

5 Explain partially order relation, Poset and Hasse diagram with example(s).

MODULE – 5

GRAPHS AND REACHABILITY

Definitions

1 Define size of a graph, adjacent vertices (nodes), finite graph, infinite graph and null graph.

2 Define isolated node, self-loop, pendent vertex, simple graph, parallel edges and multi-graph.

3 Define undirected graph, directed graph, initiating and terminating nodes, weighted graph, complete graph, path, elementary cycle, walk and cycle.

4 Define degree of a vertex, indegree of a vertex, outdegree of a vertex, total degree of a vertex, subgraph, vertex disjoint and edge disjoint subgraphs, complement of a subgraph.

5 Explain the regular graph, bipartite graph and complete bipartite graph with example(s).

Hand Shaking theorem

1 Does there exist a graph with 20 edges and each vertex of degree 3?

2 Find the number of edges in G if it has 6 vertices each of degree 2.

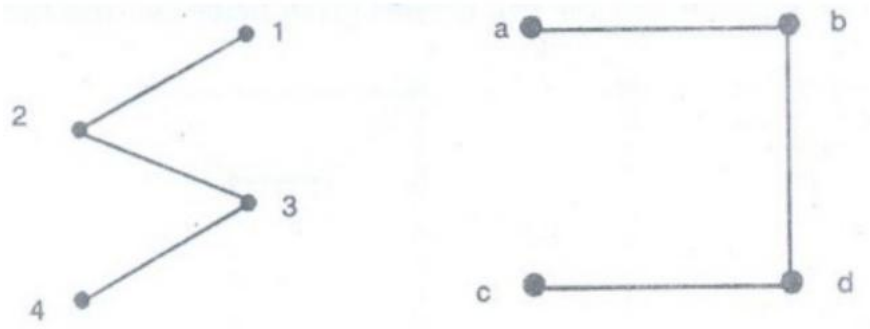
3 In a graph, each vertex has degree 3 and the total number of edges is 9. Find the number of vertices in the graph.

4 A graph G has 15 edges, 3 vertices of degree 4 and other vertices of degree 3. Find the number of vertices in G.

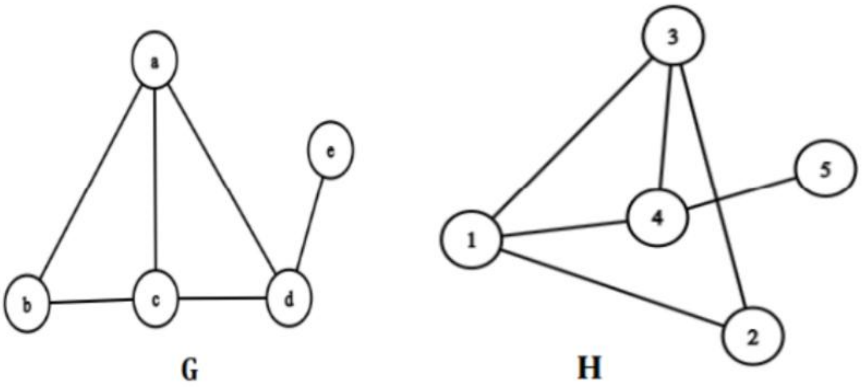
5 How many nodes are necessary to construct a graph with exactly 8 edges in which each node is of degree 2?

Isomorphic Graphs

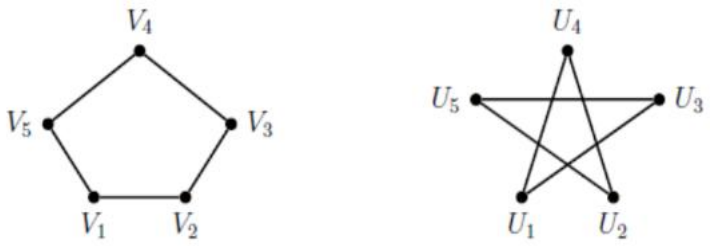
1 Define Isomorphic graphs. Check whether the following graphs are isomorphic or not.



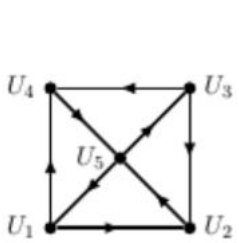
2 Check whether the following graphs are isomorphic or not.



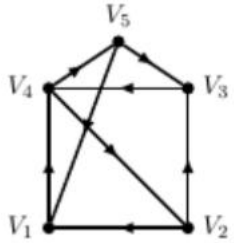
3 Check whether the following graphs are isomorphic or not.



4 Check whether the following graphs are isomorphic or not.

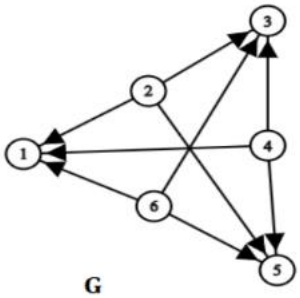


$G(V, E)$

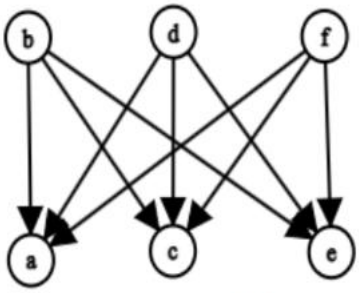


$H(V, E)$

5 Determine whether the following graphs are isomorphic.



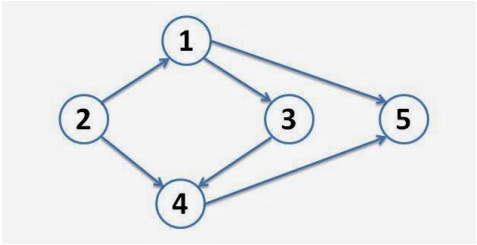
G



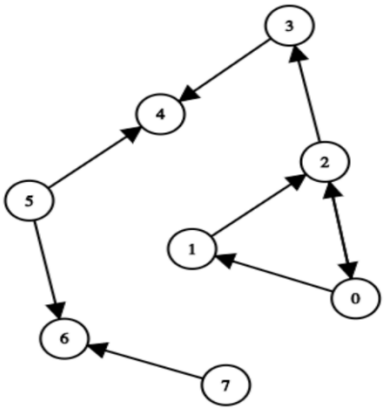
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Reachable set of each vertex

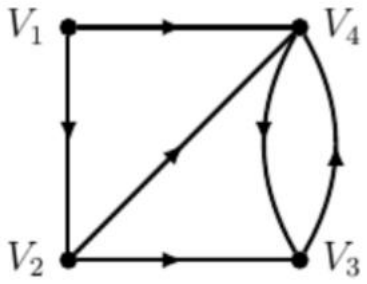
1 Find reachable set of each vertex of the given digraph.



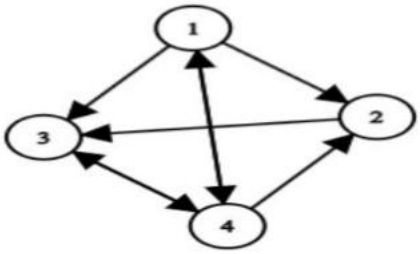
2 Find reachable set of each vertex of the given digraph.



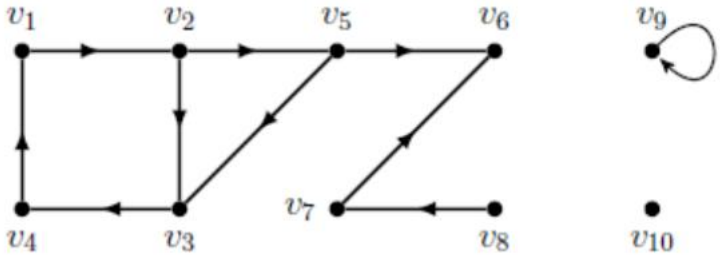
3 Find reachable set of each node of the given digraph.



4 Find reachable set of each vertex of the given digraph.



5 Find reachable set of each node of the given digraph.



IMPORTANT TOPICS

MODULE – 2

SET THEORY

- Inclusion - Exclusion Principle

COUNTING AND COMBINATORICS

- Combinations without repetition (with restrictions)

MODULE – 3

FUNCTIONS

- Composition of two or three functions

MODULE – 4

RELATIONS

- Equivalence relation
- Warshall's Algorithm to find transitive closure of a relation

PARTIAL ORDERING

- Hasse Diagram

MODULE – 5

GRAPHS AND REACHABILITY

- Definitions
- Handshaking lemma (theorem)
- Isomorphic Graphs
- Reachable set of each vertex